The EXPRESS-G language

EXPRESS-G

EXPRESS-G is a graphical subset of EXPRESS lexical language and has evolved from a variety of sources.

10/88

EXPRESSAM (Bruce Lownsbery, Lawrence Livermore Lab) based on NIAM.

2/89

GREXP (Peter Wilson, GE) BLA (boxes, lines, annotation) with multiple levels of abstraction and any amount of text.

11/89

ExpressGraph (Curt Parks, NIST & John Zimmerman, Allied Bendix) simplified version of EXPRESSAM.

2/90

EXPRESS-G (EXPRESS Committee)

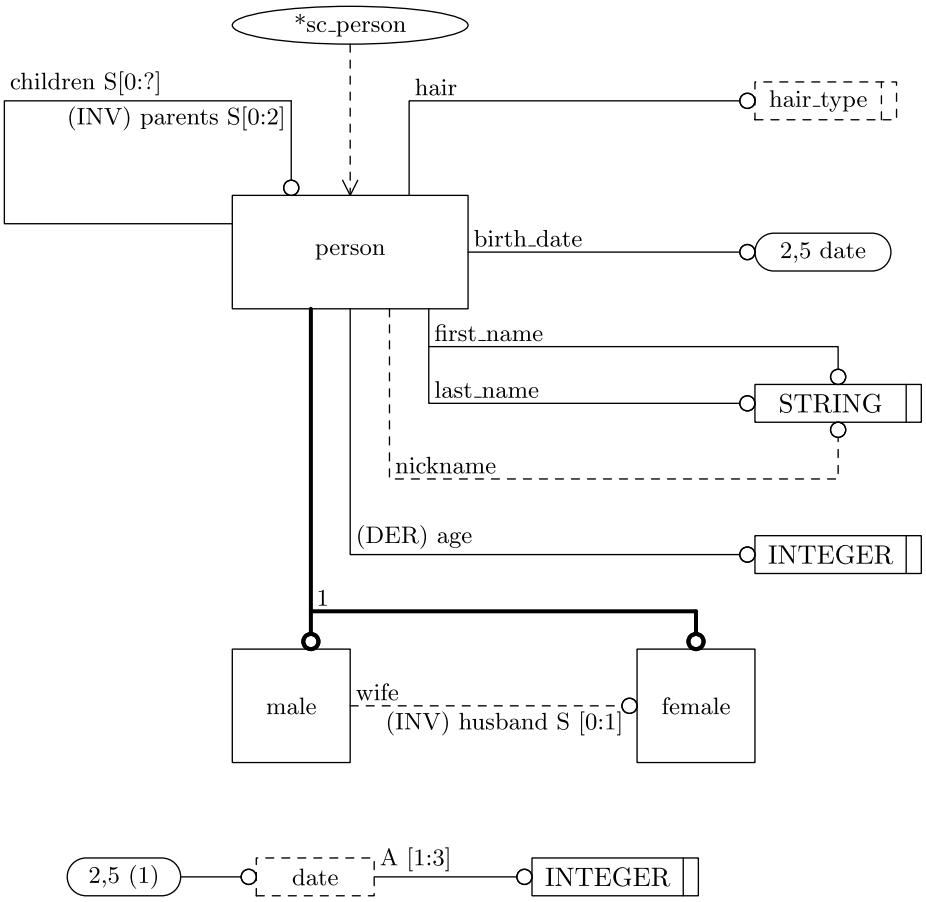
EXPRESS-G

- Drafted by EXPRESS Committee in January 1990
- A combination of ExpressGraph and GREXP.
- A *subset* of EXPRESS.
- Modeling of Entity, Attribute, Cardinality, Supertype, Type and Schema.
- Limited levels of abstraction.
- No ISO ballot issues in mid 1990 (Informative Annex).
- February 1991 — Normative Annex to EXPRESS document and extended.
- Same standardization status as EXPRESS.

EXPRESS-G Design Goals

- Subset of EXPRESS.
- Intuitive understanding of diagrams.
- Support different levels of abstraction.
- Mechanism for multi-page models.
- Minimal graphic capabilities needed.
- Potential automatic EXPRESS-G display of EXPRESS models.

EXPRESS-G Example Model



Definition Symbols

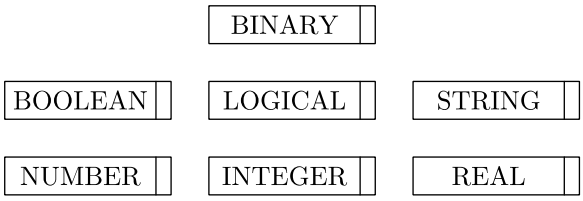


Figure 1. BASE TYPES

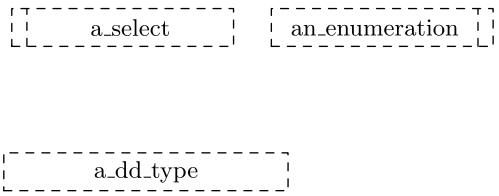


Figure 2. DEFINED TYPES

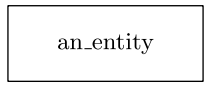


Figure 3. ENTITY

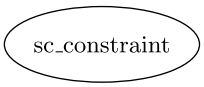


Figure 4. SUBTYPE_CONSTRAINT

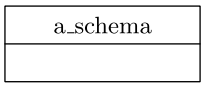


Figure 5. SCHEMA

Relationship Symbols

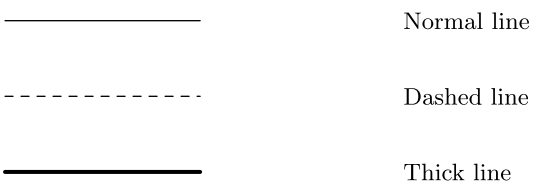


Figure 6. LINE STYLES

Composition Symbols

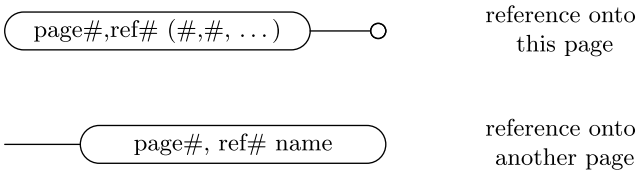


Figure 7. PAGE REFERENCES

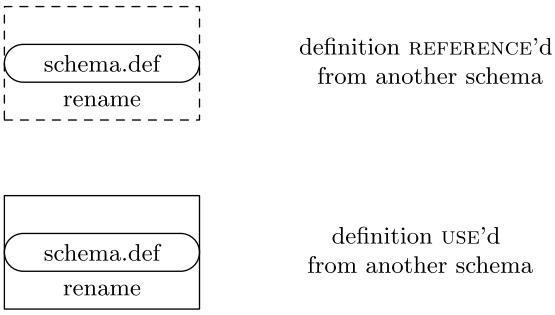
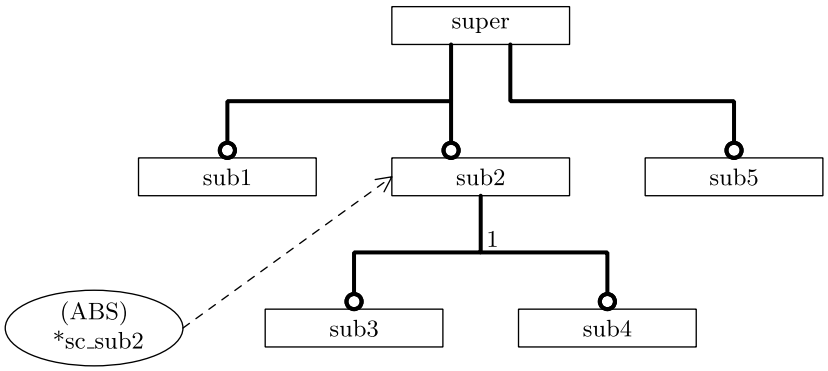
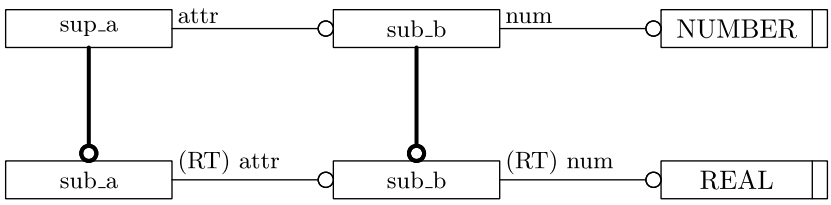


Figure 8. INTER-SCHEMA REFERENCES

A Supertype Tree



Retyping attributes



Partial and Complete Entity Models

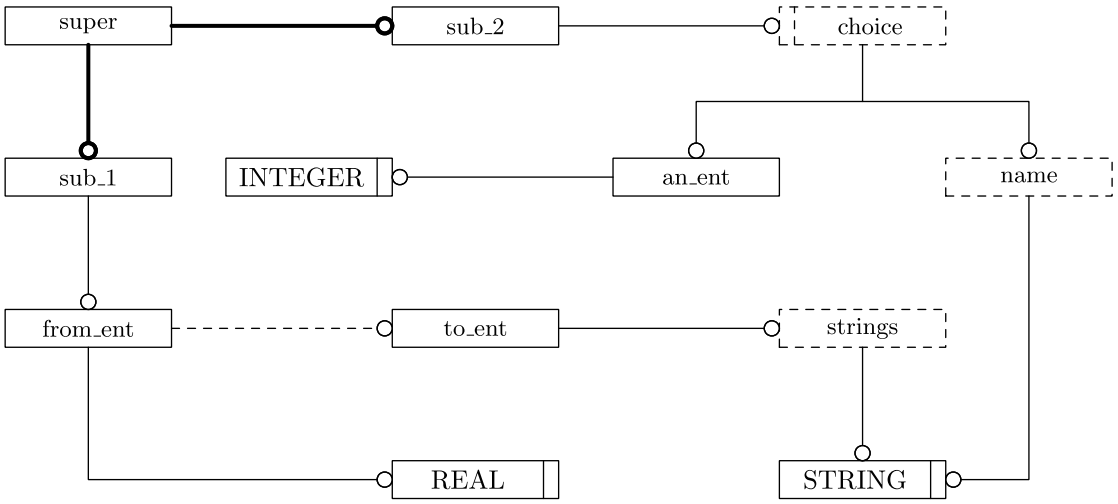


Figure 9. PARTIAL ENTITY LEVEL MODEL

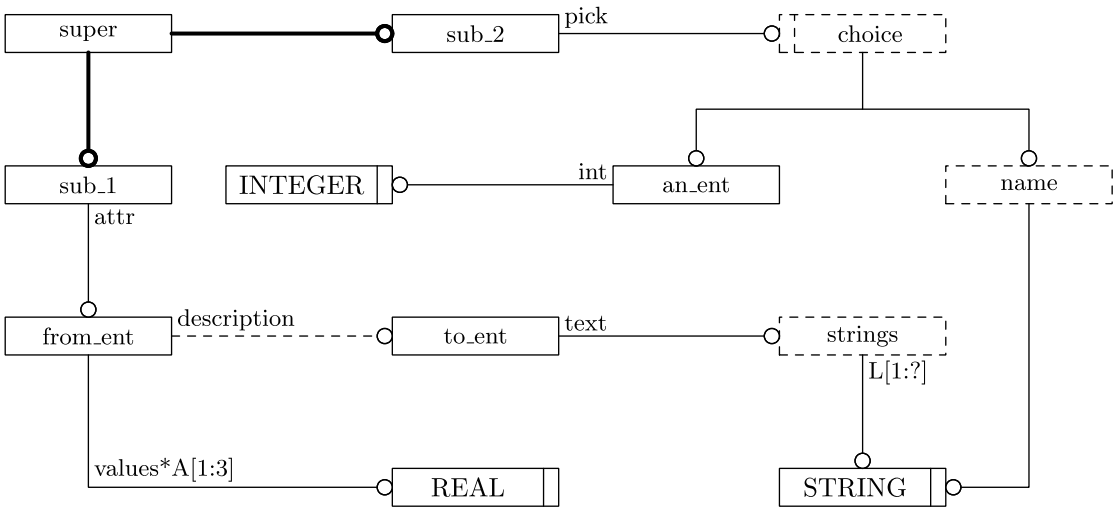


Figure 10. COMPLETE ENTITY LEVEL MODEL

Schema and Entity Models

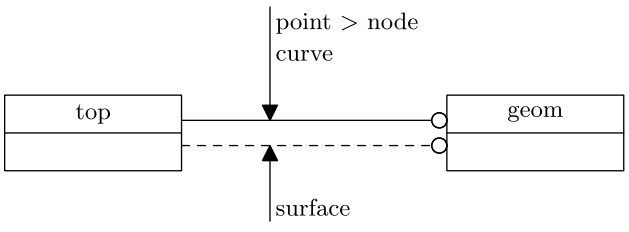


Figure 11. SCHEMA LEVEL MODEL

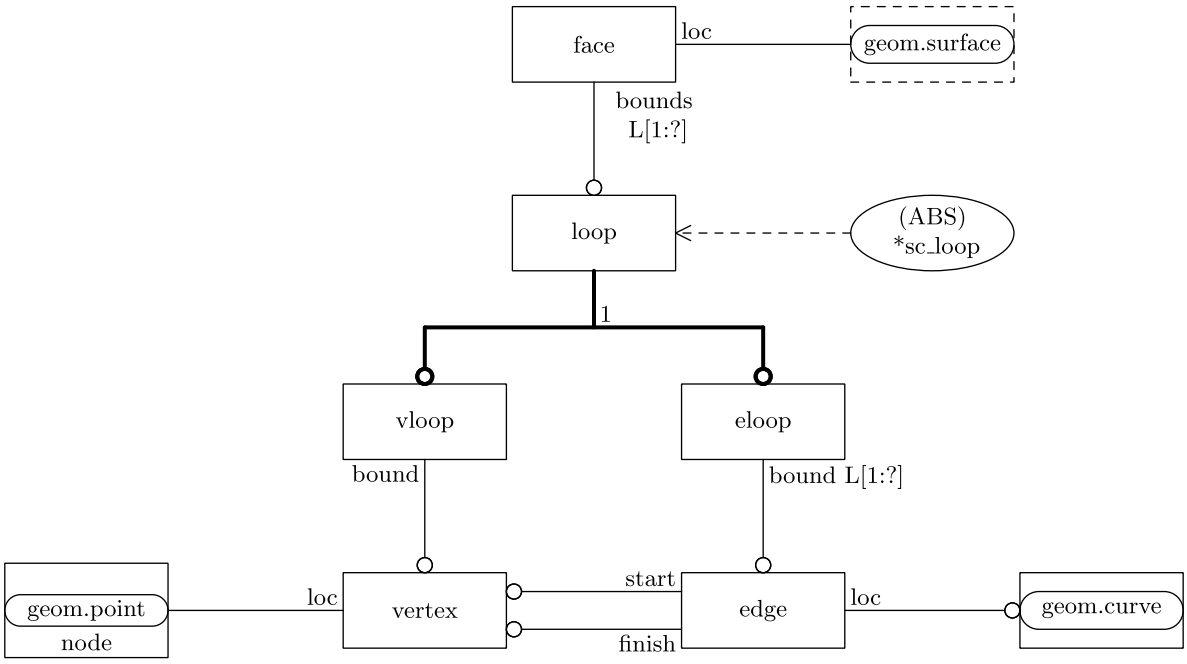
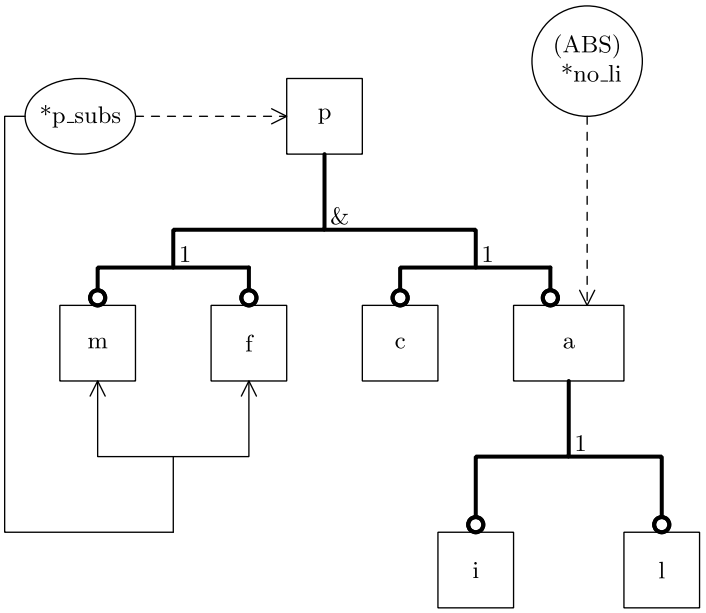


Figure 12. ENTITY LEVEL MODEL

Subtype constraints



Usage

- Graphical display of EXPRESS models.
- Stand-alone information modeling iconic language.

- Model display at varying levels of abstraction.
- Model display at varying levels of granularity.